

```
[root@lustrerclient1 conf]# cat core-site.xml
<?xml version="1.0"?>
<configuration>
<!-- Put site-specific property overrides in this file. -->

</configuration>
<property>
<name>fs.default.name</name>
<value>hdfs://lustrerclient1:8020</value>
</property>
<property>
<name>fs.file.impl</name>
<value>org.apache.hadoop.fs.LocalFileSystem</value>
</property>
<name>hadoop.tmp.dir</name>
<value>/mnt/lustre/hadoop_tmp</value>
</property>
</configuration>
```

Figure 7: core-site.xml

```
[root@lustrerclient1 conf]# cat mapred-site.xml
<?xml version="1.0"?>
<configuration>
<!-- Put site-specific property overrides in this file. -->

</configuration>
<property>
<name>mapred.job.tracker</name>
<value>hdfs://lustrerclient1:8020</value>
<description>Name and port that the MapReduce jobtracker runs at.</description>
</property>
</configuration>
```

Figure 8: mapred-site.xml

MDS. Use the following commands, specifying the IP address of the MDS server:

```
#mount -t lustre 192.168.1.185@tcp0:/lustre /mnt/lustre
```

You can use the *lfs* utility to manage the entire file system information at the client system (as shown in Figure 6).

The figure shows that the overall file system size of */mnt/lustre* is around 70GB. Striping of data is an important aspect of the scalability and performance of the Lustre file system. The data gets striped over the blocks of multiple OSTs. The stripe count can be set on a file system, directory or file level.

You can view the striping details by using the following command:

```
[root@lustrerclient1 ~]# # fs getstripe /mnt/lustre
/mnt/lustre
Stripe_count: 1 stripe size: 1048576 stripe_offset: -1
```

The Lustre set-up is now ready. It's time to run Hadoop over Lustre instead of HDFS.

I assume that Hadoop is already running with one name node and four data nodes.

On the master node, let's perform the following file configuration changes. Open the files */usr/local/hadoop/conf/core-site.xml* and */usr/local/hadoop/conf/mapred-site.xml* in any text editor and make changes as shown in Figures 7 and 8.

On every slave node (data nodes), let's perform the configuration changes as done above.

```
[root@lustrerclient1 hadoop]# bin/start-mapred.sh
Starting jobtracker, logging to /usr/local/hadoop/libexec/../logs/hadoop-root-jps
lustrerclient1:8020
10.46.214.189: starting tasktracker, logging to /usr/local/hadoop/libexec/../logs
hadoop-root-tasktracker-lustrerclient1.com
```

Figure 9: Starting mapred service

```
[root@lustrerclient1 hadoop]# bin/hadoop jar hadoop-examples-1.1.1.jar wordcount
/mnt/lustre/oboo /mnt/lustre/oboo default
10/03/21 06:49:00 INFO util.NativeCodeLoader: Loaded the native hadoop library
10/03/21 06:49:00 INFO InputFormatBase: Total input paths to process is 1
10/03/21 06:49:00 WARN MapReduceShuffle: Shuffle native library not loaded
10/03/21 06:49:00 INFO mapred.JobClient: Running job: job_0010001091_0001
10/03/21 06:49:01 INFO mapred.JobClient: map 0% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 1% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 2% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 3% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 4% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 5% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 6% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 7% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 8% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 9% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 10% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 11% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 12% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 13% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 14% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 15% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 16% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 17% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 18% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 19% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 20% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 21% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 22% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 23% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 24% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 25% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 26% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 27% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 28% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 29% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 30% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 31% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 32% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 33% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 34% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 35% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 36% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 37% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 38% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 39% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 40% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 41% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 42% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 43% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 44% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 45% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 46% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 47% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 48% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 49% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 50% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 51% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 52% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 53% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 54% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 55% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 56% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 57% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 58% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 59% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 60% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 61% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 62% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 63% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 64% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 65% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 66% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 67% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 68% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 69% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 70% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 71% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 72% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 73% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 74% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 75% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 76% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 77% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 78% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 79% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 80% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 81% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 82% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 83% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 84% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 85% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 86% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 87% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 88% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 89% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 90% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 91% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 92% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 93% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 94% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 95% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 96% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 97% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 98% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 99% reduce 0%
10/03/21 06:49:01 INFO mapred.JobClient: map 100% reduce 0%
```

Figure 10: Output for hadoop word count

Once the configuration is done, we are good to start the Hadoop-related services.

On the master node, run the *mapred* service without starting HDFS (since we are going to use only Lustre) as shown in Figure 9.

You can ensure the service runs through the *jps* utility as follows:

```
[root@lustrerclient1 ~]# jps
20112 Jps
15561 JobTracker
[root@lustrerclient1 ~]#
```

Start the *tasktracker* on the slave nodes through the following command:

```
[root@lustrerclient2 ~]# bin/hadoop-daemon.sh start tasktracker
```

You can now just run a simple *hadoop* word count example (as shown in Figure 10). 

References

- [1] http://hadoop.apache.org/docs/r1.1.1/mapred_tutorial.html
- [2] http://wiki.lustre.org/index.php/Configuring_the_Lustre_File_System
- [3] http://wiki.lustre.org/images/1/1b/Hadoop_wp_v0.4.2.pdf

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